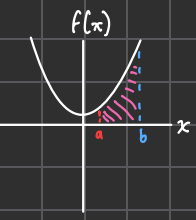


11.1: Integration as a reverse to differentiation:

If $\frac{d}{dx}(ax^n) = anx^{n-1}$

$\Rightarrow$ The reverse process is $\int kx^n\,dx = \frac{kx^{n+1}}{n+1} + C$, where C is the constant of integration, creating families of graphs.

11.2 & 11.3: Finding Areas

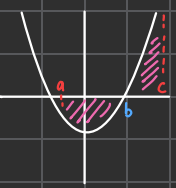


$f(x) = x^2$

$\Rightarrow \int f(x)\,dx = \frac{x^3}{3} + C$ INDEFINITE INTEGRATION

$\Rightarrow \int_a^b f(x)\,dx = \left[\frac{x^3}{3}\right]_a^b = \frac{b^3}{3} - \frac{a^3}{3}$ DEFINITE INTEGRATION

This is the area bounded by lines $x=a$ and $x=b$



$f(x) = x^2 + 2x + 1$

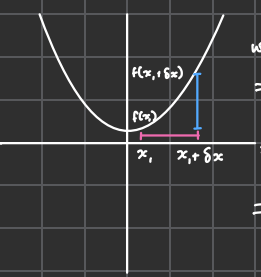
TOTAL AREA: (DO THIS UNLESS ASKED OTHERWISE)

$A = \left|\int_a^b f(x)\,dx\right| + \int_b^c f(x)\,dx$ ($\int_a^b f(x)\,dx$ is negative, hence take the abs. value of that region)

NET AREA: (VECTOR QUANTITIES; IE DISPLACEMENT OR VELOCITY)

$A = \int_a^c f(x)\,dx$

Integration can be applied to many circumstances of kx^n, but not $n=-1$ ($\int x^{-1}\,dx = \ln|x| + C$)



where δx is a very small change in x

$\Rightarrow A_1 = \delta x\, f(x) = y\,\delta x$

$\sum_{x=a}^{b} A_k = \sum_{x=a}^{b} y\,\delta x$ (y remains the same as $f(x+\delta x)$ is so small)

$\Rightarrow \sum_{x=a}^{b} y\,\delta x = \int_a^b y\,dx$